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10 **Agency enhances episodic memory across the broader experience**

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ABSTRACT

Agency, the ability to exert control over one's own experience, enhances memory relative to passive observation. However, it remains unclear whether agency only benefits information that one directly acts on, or if its benefits extend to the fidelity that the broad experience is represented in memory. Across two experiments, participants encoded objects and their locations while exploring a series of rooms, either controlling object placements themselves (Active) or observing the same sequence of events without control (Passive). Memory was assessed using the Mnemonic Similarity Task (MST), along with tests of object-spatial placement and descriptive recall. Critically, mnemonic discrimination required retaining perceptual details beyond what the object-spatial placement task demanded. In Experiment 1, agency enhanced basic object recognition, mnemonic discrimination, and object description recall, but not object-spatial placement. Experiment 2 used a within-subjects design, a recognition-based spatial test with lower retrieval demands, and attention ratings collected throughout encoding. Agency again enhanced mnemonic discrimination, and its benefits extended to object-spatial placement and recall of both object and room descriptions. These effects remained after accounting for self-reported attention during encoding. Together, the findings show that agency enhances mnemonic discrimination on the MST and suggests that its benefits extend beyond information falling within the scope of participants' direct control, influencing both the detail of individual object representations and memory for the broader context in which they were encountered.

Keywords: *Agency, episodic memory, pattern separation, mnemonic discrimination, associative memory*

1. INTRODUCTION

In everyday life, many experiences involve agency. Agency allows individuals to exert control over the experience, but its influence does more than simply determine the outcome of the experience. Consider completing a jigsaw puzzle: The person placing the pieces selects among alternatives, tests possibilities, and corrects mistakes, whereas an observer only watches the same events occur. Agency likely involves mental processes that change how the experience unfolds for an individual, potentially affecting the way they engage with and learn from what occurs. Notably, prior studies have shown that memory is enhanced by agency relative to passive observation (Bridge et al., 2017; Chrastil & Warren, 2012; DuBrow et al., 2019; Hon & Yeo, 2021; Markant, 2020; Markant et al., 2014; Murty et al., 2015; Ruiz et al., 2023; Voss et al., 2011a, b).

However, much of this work has focused on memory for individual items or the specific feature shaped by a participant's actions. Less is known about how agency affects associative memory, including how items are bound to the contexts in which they occurred, and, to our knowledge, no prior work has tested how agency influences memory for more peripheral information, beyond the scope of direct control. There is reason to suspect that agency may also influence episodic memory more broadly. Beyond supporting associations, episodic memory enables the formation of highly distinct representations to guard against interference. These two functions, associative binding and pattern separation, rely heavily on the hippocampus.

Pattern separation is the process by which overlapping experiences are represented as distinct and nonoverlapping memory representations (McClelland et al., 1995; Norman & O'Reilly, 2003; Yassa & Stark, 2011). This process reduces interference among similar experiences and preserves the specific details that distinguish one memory from another. Pattern separation has often been assessed behaviorally in humans using the Mnemonic Similarity Task

(MST), a recognition paradigm designed to tax high-fidelity memory representations (Stark et al., 2013, 2019). Typical MST paradigms include an incidental encoding phase followed by a surprise recognition test. During the test, participants are shown exact repetitions of studied objects, novel objects, and lures that are perceptually similar to studied objects. Correctly identifying a lure as similar requires maintaining highly overlapping representations as distinct, and lure discrimination is therefore used as a behavioral index of pattern separation.

Another important function of the hippocampus is binding, through which an item is integrated with the context where it was encountered (Cohen & Eichenbaum, 1993; Davachi, 2006; Ranganath, 2010). The hippocampus is broadly sensitive to context, though there is particularly strong evidence for hippocampal representation of space, including the locations of objects within a room (O'Keefe & Nadel, 1978). Binding therefore supports relational memory by preserving not only the individual elements of an experience, but also the relationships among them. Together, pattern separation and the relational binding processes capture different aspects of memory, with the former preserving the specific features that distinguish similar items and the latter connecting those items to the contexts in which they occurred.

Given these multiple functions of the hippocampus, it is unclear how agency influences different aspects of memory. Agency may improve basic recognition without enhancing the fidelity of an item or the associations formed between an item and its context. Alternatively, agency's benefits may extend to enhance other aspects of an experience as well. Identifying which aspects of memory benefit from agency is therefore important for understanding both the nature of the effect and the mechanisms that may support it.

One study directly compared memory for multiple features of a single experience (Houser et al., 2022). Participants in an Active condition explored rooms in a text-based game, choosing

1 which objects to examine and in what order while searching for a hidden key. Participants in a
 2 Passive condition encountered the same objects, rooms, and sequence but had no control over the
 3 exploration. Across two samples, agency improved memory for the order of exploration but not
 4 for object locations or descriptions (Houser et al., 2022). These findings were interpreted as
 5 evidence for a goal-relevance account because the benefit emerged only for temporal order, the
 6 feature most directly shaped by participants' choices. Under this account, agency selectively
 7 strengthens information acted on that is relevant to a participant's goal.

8 In contrast, other work suggests that agency may enhance memory more broadly. When
 9 participants viewed objects and their locations through a moveable window that they either
 10 controlled or watched follow the same path, volitional control improved both recognition of the
 11 objects and memory for their locations (Voss et al., 2011a). Neither benefit emerged in patients
 12 with hippocampal damage, suggesting that the hippocampus is important for the memory benefits
 13 of agency. A broader memory benefit is also grounded in the proposal that agency promotes an
 14 interrogative motivational state (Murty & Adcock, 2017). In this state, one actively explores the
 15 environment and resolves uncertainty in pursuit of a goal. This form of motivation is proposed to
 16 engage hippocampal and dopaminergic systems that support relational encoding. According to this
 17 account, agency should enhance not only the targets of goal pursuit, but also other features and
 18 associations encountered during the experience (Murty & Adcock, 2017). Agency may therefore
 19 shape memory beyond the information a participant directly controls.

20 Together, prior work provides competing ideas about how agency influences memory.
 21 Agency may selectively strengthen the feature shaped by a participant's choices, or it may more
 22 broadly enhance the items and contexts that make up an experience. Here, we tested whether
 23 agency benefits only the information controlled or more broadly influences multiple features of

1 how an experience is represented in memory. Participants played or watched a sorting game, where
 2 they encoded objects and locations while exploring a series of rooms. In each room, participants
 3 placed a set of objects into their correct locations through trial and error, advancing only after
 4 every object had been correctly placed. In the Active condition, participants controlled these
 5 placements, while in the Passive condition, participants observed the Active participant's
 6 gameplay. Critically, Passive participants experienced the same objects, rooms, and sequence of
 7 events, but without control. After encoding, participants completed several memory tasks,
 8 consisting of the MST, an object-spatial placement test, and free recall of object and room
 9 descriptions. These measures were selected because the two conditions make the same prediction
 10 for information participants directly control but diverge for other features encountered during
 11 active exploration.

12 Specifically, spatial placement of objects was the feature of the task that participants
 13 actively controlled, so an agency benefit on this task would be consistent with both stronger item-
 14 context binding and the preferential strengthening of controlled information. Critically, the MST
 15 provided a more adjudicating test by assessing whether agency enhances mnemonic discrimination
 16 for objects whose perceptual features participants did not directly control, a dimension not
 17 examined in prior work. Participants acted on the objects by placing them, but they did not control
 18 the specific perceptual features that distinguished each object from similar lures. Placing an object
 19 required only identifying it well enough to select its location, whereas discriminating that object
 20 from a lure required retaining perceptual detail beyond what the task itself demanded. Thus, an
 21 agency benefit of lure discrimination would suggest that its effects extend beyond the information
 22 directly shaped through action. Additionally, recall of room descriptions provided a unique test
 23 where neither condition of participants physically acted upon the given information. Together,

these measures allowed us to test whether agency selectively enhances controlled information or more broadly benefits information that was not directly acted upon.

Experiment 1 used an in-person, between-subjects design. Agency enhanced object recognition, mnemonic discrimination, and recall of object descriptions, but not object-spatial placement memory. This pattern ran counter to the goal-relevance account and also did not provide clear support for the broader relational account, which likewise predicted enhanced item-context binding. One possibility is that allowing participants to freely place the objects without any cue imposed substantial retrieval demands, reducing its sensitivity to differences between conditions. Given that Experiment 1 could not determine whether agency selectively enhances goal-relevant information or produces a broader enhancement of memory, Experiment 2 was designed to adjudicate between these two accounts.

For Experiment 2, we used a within-subjects design, a recognition-based object-spatial placement test with lower retrieval demands, and also added free recall of room descriptions, which provided a stronger test of a broad benefit because participants never acted upon this information. Agency again enhanced mnemonic discrimination, and the benefit extended to object-location memory and recall of both object and room descriptions. These effects remained even after including attention ratings collected after each encoding block, indicating that they were not reducible to differences in attention during active encoding. Together, agency strengthened the information participants controlled but also reliably enhanced memory more broadly. Its benefits reached the perceptual details that distinguished similar objects and the room descriptions participants never acted upon, extending beyond the placements their choices determined.

2. EXPERIMENT 1

2.1 MATERIAL AND METHODS

2.1.1 *Transparency and Openness*

Materials, primary data, and analysis scripts are publicly available on [OSF Framework](#) and [Github](#). No aspect of the experiment was preregistered.

2.1.2 *Participants*

Ninety participants were recruited from a pool of undergraduate students at Washington University in St. Louis and received course credit for their participation. Inclusion criteria required that participants have fluent English proficiency and have normal or corrected-to-normal vision. Using a between-subjects experimental design, participants were randomly assigned to one of two conditions: Active or Passive. Participants in the Active condition actively completed the encoding portion of the experiment while participants in the Passive condition observed a yoked replay of an Active participant's gameplay. The yoking method ensured identical visual exposure of both conditions to the game task and formed distinct Active-Passive yoked pairs. Two Active participants were excluded due to incomplete yoked pairings, yielding a final sample of 44 Active ($M = 19.311$, $SD = 1.202$, range = 18-22; 30 female) and 44 Passive participants ($M = 19.150$, $SD = 1.167$, range = 18-21; 25 female).

2.1.3 *Procedure*

Participants completed the in-person study in a single session. Following informed consent and demographic forms, all participants completed a brief example task of the Encoding Phase to familiarize themselves with the computer controls. Briefly, participants were presented a Visuospatial Game involving a room with a set of 3 objects. Participants were instructed to drag and drop each object to a predetermined target location and received feedback indicating whether

each placement was correct. If any of the objects were incorrectly placed, participants repeated the trial until all objects were placed correctly. Participants then completed the Encoding Phase where participants were presented the Visuospatial Game to either complete (Active) or observe (Passive). Following the Encoding Phase, all participants were presented with several memory tests in the Retrieval Phase including a Mnemonic Similarity Task (MST), Object Description Test, and Object-Spatial Placement Test. Participants then completed a post-task survey assessing their perceived sense of agency and attention and were debriefed of the experimental manipulations.

2.1.3.1 Encoding Phase

Participants were introduced to the Visuospatial Game consisting of an Object Description Task and Object-Spatial Placement Task. Critically, Active participants played the Visuospatial Game, while Passive participants viewed a yoked replay of an Active participant's gameplay. The Visuospatial Game included ten rooms, each containing a set of 5 objects, yielding a total of 50 encoded objects.

Object Description Task. Within each room, participants were instructed to learn a description of each object's special capability (e.g., "*This object has unlimited memory capacity*"), which were unnatural traits to invalidate guessing during recall.

Object-Spatial Placement Task. Five objects appeared in randomized locations within each room, alongside five predetermined target locations. Participants dragged and dropped each object to a target location and received feedback indicating whether each placement was correct. Participants repeated the trial until all objects were placed correctly.

2.1.3.2 Retrieval Phase

While the Encoding Phase consisted of participants either playing (Active) or observing (Passive) the game, all participants completed identical Retrieval Phase consisting of 3 memory tasks assessing object recognition, object description memory, and object-spatial placement memory.

Mnemonic Similarity Task (MST). The Mnemonic Similarity Task (MST) is a well-established recognition memory paradigm designed to assess both basic recognition memory and fine-grained mnemonic discrimination (Stark et al., 2019). Participants were presented an object at the center of the screen and judged whether the object was “Old” (an original target seen during the Encoding Phase), “Similar” (a highly similar but non-target lure), or “New” (a foil object not seen during the Encoding Phase). The task consisted of 25 original Target objects, 25 similar Lures, and 25 novel Foils, resulting in 75 total responses. From these responses, we calculated Basic Object Recognition (d') and the Lure Discrimination Index (LDI; Stark et al., 2019).

Basic Object Recognition (d'). We first assessed Basic Object Recognition by calculating d' , which is a measure of recognition sensitivity, reflecting a participant’s ability to discriminate previously seen targets from novel foils, independent of response bias. The d' value was calculated as:

$$d' = z(\text{Hit Rate}) - z(\text{False Alarm Rate})$$

where the Hit Rate is the proportion of Targets correctly identified as “Old,” and the False Alarm Rate is the proportion of Foils incorrectly identified as “Old”. Higher d' values indicate greater recognition memory sensitivity.

Lure Discrimination Index (LDI). Next, we assessed participants’ ability to discriminate the highly similar Lures from previously presented Targets. This measure is thought to reflect hippocampal pattern separation processes, indexing the extent to which objects with overlapping

perceptual features are encoded as distinct representations. We calculated the Lure Discrimination Index (LDI) as:

$$LDI = p(\text{"Similar"}|Lure) - p(\text{"Similar"}|Target)$$

where the proportion of Targets incorrectly identified as “Similar” is subtracted from the proportion of Lures correctly identified as “Similar.” Higher LDI values reflect better mnemonic discrimination of perceptually overlapping items.

For each object identified as “Old”, regardless of accuracy, participants proceeded to subsequent memory tasks for that object, including the Object Description Test and Object-Spatial Placement Test.

Object Description Recall. For objects identified as “Old” in the MST, participants were then presented a blank textbox and prompted to type what they remembered about the object’s special capability as presented during the Visuospatial Game. Responses were scored with semantic sentence embeddings using Google’s Universal Sentence Encoder (USE; Cer et al., 2018). Each participant’s response was compared with the correct description using cosine similarity between sentence-level embedding vectors. Higher values indicated greater semantic similarity between the recalled and correct descriptions.

Object-Spatial Placement Test. Participants were then presented all rooms encountered during the Encoding Phase and asked to select the room the object belonged to. The object was then presented in the selected room, and the participants were instructed to drag the object to its correct target location from the Visuospatial Game. Spatial memory accuracy was quantified as the Euclidean distance, in pixels, between the participant’s response coordinate and the correct target coordinate, such that lower scores indicate more accurate spatial memory.

Post-Task Questionnaire. Immediately after the Retrieval Phase, participants completed a Post-Task Questionnaire which was a single, self-reported assessment measuring attention, sense of agency, confidence, and strategy (Table 1). Participants rated their level of agreement with a series of statements using a Likert scale (1 = Strongly Disagree, 6 = Strongly Disagree).

2.1.4 Analyses

Given that Experiment 1 used a between-subjects design, each dependent measure was analyzed with ordinary least-squares (OLS) regression. We fit a model that included Agency (Active versus Passive) as the predictor in the following model:

$$\text{Dependent Variable} \sim \text{Agency}$$

The dependent measures included basic recognition (d') and the Lure Discrimination Index (LDI) from the MST, description accuracy (cosine similarity) from the Object Description Test, and spatial placement error (Euclidian distance in pixels) from the Object-Spatial Placement Test.

2.2 RESULTS

2.2.1 Basic Object Recognition.

We first assessed basic recognition sensitivity (d') for objects using responses from the Mnemonic Similarity Task (MST). Basic recognition measured participants' ability to identify previously seen Target objects from novel Foil objects. An OLS regression revealed a significant effect of Agency ($b = -0.793$, $SE = 0.174$, $t(86) = -4.553$, $p < .001$, $95\% \text{ CI} = [-1.139, -0.447]$), with Active participants demonstrating higher recognition sensitivity ($M = 2.284$, $SD = 0.690$) than Passive participants ($M = 1.491$, $SD = 0.927$).

2.2.2 Lure Discrimination Index (LDI).

Next, we assessed participant's ability to distinguish previously seen Target items from perceptually similar Lure items using the Lure Discrimination Index (LDI). An OLS regression revealed a significant effect of Agency ($b = -0.206$, $SE = 0.068$, $t(86) = -3.026$, $p = .003$, 95% CI = $[-0.342, -0.071]$), with Active participants ($M = 0.244$, $SD = 0.321$) demonstrating higher lure discrimination ability than Passive participants ($M = 0.037$, $SD = 0.318$). Together, results from the MST measures indicate that the Agency enhances recognition memory for correctly identifying objects.

2.2.3 *Object-Spatial Placement Test.*

We assessed spatial memory using spatial placement error, which is calculated as the Euclidean distance between each object location placement and the object's correct target location. Lower distance values indicated more accurate spatial memory. An OLS regression revealed no significant effect of Agency ($b = -1.718$, $SE = 5.799$, $t(86) = -0.296$, $p = .768$, 95% CI = $[-13.246, 9.810]$), with Active ($M = 150.805$, $SD = 23.816$) and Passive ($M = 149.087$, $SD = 30.206$) participants showing comparable placement accuracy.

2.2.4 *Object Description Recall.*

Object-description memory was assessed using semantic similarity scores comparing each participant's recalled description to the original encoded description. Higher semantic similarity values indicated greater overlap between the recalled and original descriptions, and therefore better object-description recall. An OLS regression revealed a significant effect of Agency ($b = -0.085$, $SE = 0.030$, $t(86) = -2.866$, $p = .005$, 95% CI = $[-0.143, -0.026]$), with Active participants demonstrating higher object-description recall ($M = 0.346$, $SD = 0.134$) than Passive participants ($M = 0.261$, $SD = 0.142$).

2.3 DISCUSSION

Here, we tested whether agency enhances memory selectively for controlled information or enhances memory more broadly. Participants completed memory tests that differed in their relevance to participants' goals during the Encoding Phase. Object-spatial placement was goal-relevant, whereas basic object recognition, mnemonic discrimination, and object description recall were not. The goal-relevance account predicted an agency benefit for spatial placement but not for perceptual recognition or descriptive recall, whereas a broad account predicted better performance among Active participants across all measures. Interestingly, we found an agency effect for object memory, including higher recognition sensitivity, better lure discriminability, and better object-description recall. Object-spatial placement performance, however, did not differ significantly between Active and Passive participants, despite being the feature participants most directly acted upon.

This pattern is informative because agency has previously been thought to selectively benefit the actively controlled dimension (Houser et al., 2022). Our Visuospatial Game was designed so that spatial placement was the most goal-relevant feature, as correct placement advanced participants through the game. Yet the agency effect emerged for object identity and recall, features less directly tied to task completion, rather than for placement itself. Notably, the benefit appeared on the measure we identified as diagnostic. Lure discrimination requires perceptual detail beyond what the task demanded, and an advantage there is difficult to attribute to the preferential strengthening of controlled information. At the same time, the results do not map cleanly onto a broad account, because the spatial placement of objects benefit that account also predicts did not emerge.

One explanation comes from the same framework that motivates the broad account. Murty and Adcock (2017) distinguish two motivational states with different mnemonic consequences. As

1 stated previously, interrogative states, characterized by open-ended exploration and the resolution
 2 of uncertainty, are proposed to support hippocampally dependent relational encoding. Imperative
 3 states, driven by proximal goal pursuit, instead favor memory for items over the associations
 4 among them. The structure of our task may have pushed Active participants toward the latter.
 5 Given that participants advanced only after every object was correctly placed, encoding proceeded
 6 under pressure to identify each object and test it against candidate locations. Agency may therefore
 7 have strengthened memory for the objects themselves more than for their precise spatial locations.
 8 This interpretation is also consistent with our finding of an object recognition benefit in the absence
 9 of a location benefit, whereas Voss and colleagues (2011a) observed both.

10 Alternatively, it is possible that the Object-Spatial Placement Test may have imposed
 11 retrieval demands high enough to obscure a spatial benefit, as participants placed objects freely
 12 without any cue. Additionally, free placement of objects is a relatively difficult task, and the
 13 difficulty of this procedure could have obscured subtle differences between the Active and Passive
 14 participants. Moreover, Active participants may have been more engaged during encoding than
 15 Passive participants. Given that attention was assessed only once at the end of the experiment, we
 16 could not determine whether the object memory benefits reflected agency itself or greater
 17 engagement during active encoding.

18 We therefore conducted Experiment 2 to address these limitations. Experiment 2 used a
 19 within-subjects design with counterbalanced blocks of Active and Passive trials during encoding,
 20 allowing us to control for individual differences in ability and engagement. Attention ratings
 21 collected after each block provided a measure of attention during encoding rather than
 22 retrospectively, and a recognition-based Object-Spatial Placement Test reduced retrieval
 23 difficulty. We also extended description recall to rooms, to assess whether the benefits of agency

AGENCY ENHANCES MEMORY ACROSS THE EXPERIENCE

1 are specific to objects or extend to contextual information participants never acted upon. Together,
2 these changes allowed us to test whether agency enhances mnemonic discrimination above and
3 beyond attention, and whether a spatial benefit would emerge when the placement test was less
4 demanding.

5

3. EXPERIMENT 2

Experiment 2 was designed to further test how agency shapes memory representations while addressing limitations of the first experiment. Specifically, we asked whether agency enhances memory for objects beyond differences in engagement and whether benefits extend to goal-relevant features when retrieval demands are reduced.

3.1. MATERIALS AND METHODS

3.1.1. Participants

Experiment 2 used a within-subjects design wherein each participant completed both Active and Passive conditions. Thirty-one participants were recruited online through Prolific. Eligibility was restricted to adults aged 18-35 years old residing in the United States, with normal or corrected-to-normal vision, and an approval rating of at least 95% and 20 or more prior submissions on Prolific. Additionally, checks consisting of arithmetic problems were embedded throughout the experiment as a priori exclusion criteria. All participants scored proficient and within the acceptable time range on the engagement checks, yielding a final sample of 31 participants ($M = 27.970$, $SD = 4.461$, range = 20-34, 17 female). Participants received monetary compensation that was consistent with Prolific's recommended rate.

3.1.2. Procedure

Participants completed the study in a single online session. After providing informed consent and demographic information, participants completed a brief Example Task to familiarize them with the task structure. Unlike Experiment 1, Experiment 2 used a within-subjects design in which all participants completed both Active and Passive conditions during the Encoding Phase. Active and Passive blocks were presented in alternating order, with the starting condition counterbalanced across participants. After each block, participants rated their attention, providing

a measure of engagement during encoding rather than a single retrospective rating at the end of the session. All participants were then presented with several memory tasks in the Retrieval Phase including a Mnemonic Similarity Task (MST), Object Description Task, Room Description Task, and Object-Spatial Placement Task, before being debriefed of the experiment design.

3.1.2.1. Encoding Phase

Participants were presented with the Visuospatial Game consisting of a Room and Object Description Task and Object-Spatial Placement Task similar to that of Experiment 1. Critically, participants completed both conditions, actively playing the game in Active blocks and observing a standardized gameplay recording in Passive blocks. The Visuospatial Game included 24 rooms total, each containing 3 objects, yielding 48 total encoded objects. Rooms were presented in alternating blocks of 4 rooms from the same condition, with the starting condition counterbalanced across participants.

Description Tasks. Participants were first shown an image of the room and a description of the room's special capability (e.g., "In this room, the food on the table gets cold, but the drinks get hot;" Room Description Task). Similarly, participants were presented three objects and instructed to learn the description of each object's special capability (Object Description Task), as in Experiment 1.

Object-Spatial Placement Task. The 3 objects appeared in randomized locations within each room, alongside 3 predetermined target locations. In Active blocks, participants dragged and dropped each object to a target location and received feedback indicating whether each placement was correct, repeating the trial until all objects were placed correctly. In Passive blocks, participants viewed a standardized recording of this gameplay.

Attention Ratings. After each block, participants rated their attention during that block on a 7-point Likert scale. These block-level ratings provided multiple condition-specific measures of attentional engagement during the Encoding Phase.

3.1.2.2. Retrieval Phase

Following the Encoding Phase, all participants completed four blocks of memory tests assessing object recognition, object-spatial placement memory, object description memory, and room description memory. Each of the 48 encoded objects appeared in either the Mnemonic Similarity Task (MST) or the Spatial Placement Task, with 24 randomly assigned to each. The order of these tests was counterbalanced across participants, followed by Object Description Recall and Room Description Recall in randomized order.

Mnemonic Similarity Task (MST). The MST followed the same procedure as Experiment 1. Briefly, participants were presented an object at the center of the screen and judged whether the object was “Old,” “Similar,” or “New.” From these responses, we calculated Basic Recognition (d') and the Lure Discrimination Index (LDI) as described in Experiment 1.

Object-Spatial Placement Test. The Spatial Placement Test was administered as a modified recognition-based spatial memory task. Participants were presented with the room in which the object was encountered during the Encoding Phase, along with 3 markers indicating the object’s possible target locations. Participants dragged the object to the marker they believed matched its correct target location. This recognition-based format differed from the free placement test used in Experiment 1, in which participants placed objects anywhere within the room background, and was used to reduce the precision demands of retrieval. Each placement was scored as correct or incorrect, and spatial memory accuracy was calculated as the proportion of objects correctly placed, separately for Active and Passive conditions. Given that each object had three

possible target locations, chance performance was 33%. Higher proportions indicate more accurate spatial memory.

Object Description Recall. The Object Description Recall followed the same procedure as Experiment 1. Participants were presented a blank textbox and prompted to type what they remembered about the object's special capability as presented during the Visuospatial Game. Responses were scored using cosine similarity between sentence-level embedding vectors (Cer et al., 2018).

Room Description Recall. The Room Description Test assessed memory for the room descriptions presented during the Encoding Phase. Participants were presented each room and prompted to type what they remembered about the room's special capability. Responses were scored using the same procedure as the Object Description Task.

Post-Task Questionnaire. Immediately after the Retrieval Phase, participants completed a Post-Task Questionnaire, similar to that of Experiment 1, containing a single, self-reported assessment measuring memory confidence and control (Table 2). Participants rated their level of agreement with a series of statements using a Likert scale (1 = Strongly Disagree, 6 = Strongly Disagree).

3.1.3. Analyses

Experiment 2 used a within-subjects design in which each participant completed Active and Passive conditions. Given that each participant contributed to both, Observations were therefore nested within participant, and each dependent measure was analyzed with a linear mixed-effects model including a random intercept for participant. For each dependent measures, we fit two models. The first included Agency (Active versus Passive) as the only fixed effect:

$$Dependent\ Variable \sim Agency + (1|Subject)$$

The dependent measures were basic recognition (d') and the Lure Discrimination Index (LDI) from the Mnemonic Similarity Task, cosine similarity from the Object Description Task and Room Description Task, and placement accuracy from the Spatial Placement Task.

Given that self-reported attention may have contributed to memory performance, the second model added Attention Rating as a fixed effect:

$$\text{Dependent Variable} \sim \text{Agency} + \text{Attention Rating} + (1|\text{Subject})$$

Attention Ratings was a continuous variable derived from a 7-point Likert scale rating collected after each encoding block and averaged within condition for each participant. Unlike Experiment 1, in which attention was assessed once at the end of the session, these block-level ratings provided a measure of attentional engagement during encoding. The attention-adjusted model examined whether condition differences in memory performance persisted after accounting for differences in attentional engagement.

3.2. RESULTS

3.2.1. Basic Object Recognition.

We first assessed basic recognition (d') for objects using responses from the Mnemonic Similarity Task (MST). A linear mixed-effects model revealed a significant effect of agency ($\beta = -0.671$, $SE = 0.200$, $z = -3.350$, $p < .001$, 95% CI = $[-1.064, -0.279]$), with Active blocks producing higher recognition sensitivity ($M = 0.854$, $SD = 0.951$) than Passive blocks ($M = 0.182$, $SD = 0.845$). This effect remained significant after including attention ratings as a covariate ($\beta = -0.840$, $SE = 0.249$, $z = -3.373$, $p < .001$, 95% CI = $[-1.329, -0.352]$). Additionally, attention Ratings did not significantly predict basic recognition performance ($\beta = 0.095$, $SE = 0.085$, $z = 1.123$, $p = .262$, 95% CI = $[-0.071, 0.262]$). Together, these results indicate that the effect of Agency on basic recognition persists independently of attentional engagement.

3.2.2. *Lure Discrimination Index (LDI).*

Next, we assessed lure discriminability using the Lure Discrimination Index (LDI). A linear mixed effects model revealed a significant effect of Agency ($\beta = -0.315$, $SE = 0.057$, $z = -5.495$, $p < .001$, 95% CI $[-0.428, -0.203]$), with higher lure discrimination for Active ($M = 0.143$, $SD = 0.215$) than Passive ($M = -0.172$, $SD = 0.363$) trials. This effect remained significant after including Attention Ratings as a covariate ($\beta = -0.278$, $SE = 0.076$, $z = -3.681$, $p < .001$, 95% CI $[-0.426, -0.130]$). Attention Ratings did not significantly predict LDI ($\beta = -0.021$, $SE = 0.028$, $z = -0.756$, $p = .450$, 95% CI $[-0.076, 0.034]$). Together, these results address the ambiguity of Experiment 1. Whereas the effect of agency may have been driven by overall attention and engagement in Experiment 1, the same effect persisted after adjustment in Experiment 2, when engagement was measured concurrently with encoding. This indicates that agency enhanced mnemonic discrimination beyond what could be explained by attentional engagement.

3.2.3. *Object-Spatial Placement Test.*

We assessed spatial memory as the proportion of objects correctly placed among three possible locations, with chance at 0.33 or 33%. A linear mixed-effects model revealed a significant effect of Agency ($\beta = -0.199$, $SE = 0.044$, $z = -4.567$, $p < .001$, 95% CI $[-0.284, -0.114]$), with more accurate placement among Active ($M = 0.570$, $SD = 0.226$) than Passive ($M = 0.371$, $SD = 0.161$) trials. This effect remained significant after including Attention Ratings as a covariate ($\beta = -0.225$, $SE = 0.055$, $z = -4.060$, $p < .001$, 95% CI $[-0.334, -0.116]$), and Attention Ratings did not predict placement accuracy ($\beta = 0.015$, $SE = 0.019$, $z = 0.783$, $p = .434$). Placement following Active trials exceeded chance ($t(30) = 5.831$, $p < .001$), whereas placement following Passive trials did not differ from chance ($t(30) = 1.305$, $p = .202$). Together, these results indicate that agency

benefited spatial memory when retrieval demands were reduced, and that this benefit was not readily attributable to attentional engagement.

3.2.4. *Object Description Recall.*

We assessed recall for object descriptions by calculating the semantic cosine similarity between each participant's recalled description and the original description. Description accuracy was higher for Active ($M = .271$, $SD = .125$) than Passive ($M = .171$, $SD = .123$) encoding trials ($\beta = -0.100$, $SE = 0.019$, $z = -5.334$, $p < .001$, 95% CI $[-0.136, -0.063]$). Importantly, the effect remained significant after adjusting for attention ($\beta = -0.096$, $SE = 0.027$, $z = -3.603$, $p < .001$), and attention did not predict description accuracy ($\beta = -0.002$, $z = -0.178$, $p = .859$).

3.2.5. *Room Description Recall.*

To test whether the description benefit was specific to objects, we included recall of room descriptions following the same procedure with participants recalling each room's special capability. Recall was more similar to the original description among Active ($M = .324$, $SD = .172$) than Passive ($M = .176$, $SD = .125$) encoding trials ($\beta = -0.148$, $SE = 0.023$, $z = -6.484$, $p < .001$, 95% CI $[-0.193, -0.103]$). This effect remained after adjusting for attention ($\beta = -0.133$, $SE = 0.032$, $z = -4.128$, $p < .001$), which did not itself predict recall ($\beta = -0.009$, $z = -0.684$, $p = .494$). The Agency benefit therefore extended beyond object descriptions to memory for room-level information.

3.3. DISCUSSION

We addressed limitations of Experiment 1 by examining whether the observed benefits of agency reflected agency itself or differences in engagement, and whether a spatial benefit would emerge when retrieval demands were reduced. Using a within-subjects design and attention ratings collected after each encoding block, we found that Active encoding enhanced all our measures,

and this effect remained after accounting for attention. Additionally, the object-spatial placement results are informative given the null effect in Experiment 1.

With a recognition-based test, Active encoding supported above-chance object-spatial placement memory, whereas placement following Passive encoding did not exceed chance. Because the same participants completed both conditions and all passed embedded engagement checks, this difference is unlikely to reflect general inattention or low effort in general. Furthermore, participants completed Passive trials at chance during the spatial recognition task in this experiment. While this clearly indicates lower performance compared to the Active trials, we are unable to adjudicate whether spatial information was encoded poorly or not at all during these trials. We note that performance was also lower in Experiment 2 than in Experiment 1 across measures, which may reflect differences in the testing environment given that Experiment 2 was conducted online. However, given that Experiment 2 used a within-subjects design, any effect of the testing environment should have influenced Active and Passive conditions similarly and is therefore unlikely to account for the agency effects. More importantly, these benefits were not limited to the object-spatial placement information participants controlled. Agency also improved memory for information participants never acted upon in either condition, including room descriptions, strongly indicating that its effects extended beyond the dimension under control.

4. GENERAL DISCUSSION

Across two experiments, we tested whether agency selectively enhances memory for the information an individual controls or more broadly shapes how an experience is represented in memory. Participants encoded objects and their locations by placing objects into their correct positions within a series of rooms through trial and error, either controlling these placements themselves (Active) or observing the same sequence of events without control (Passive). We assessed memory along dimensions that varied in their relationship to participants' actions during encoding including object placement in the room, basic object recognition, mnemonic discrimination of objects, and recall of object and room descriptions. Critically, object-spatial placement was the only measure that assessed information participants directly controlled. The remaining measures assessed information their actions did not determine, including perceptual details the task never demanded and descriptive information they never acted upon.

In Experiment 1, agency enhanced object recognition, mnemonic discrimination, and object description recall, but not object-spatial placement. In Experiment 2, which used a within-subjects design, a recognition-based object-spatial placement test, and attention ratings collected throughout encoding, agency enhanced every measure, including object-spatial placement and recall of room descriptions, and each benefit persisted after accounting for attention. Together, these findings indicate that agency strengthened the information participants controlled but also enhanced memory more broadly, extending to features of the experience participants did not directly shape through action.

These findings speak directly to two accounts of how agency shapes memory. Under a goal-relevance account, agency preferentially strengthens the information that is determined through one's actions or relevant to one's goals (Houser et al., 2022). Under a broader account,

agency promotes a motivational and exploratory state that enhances encoding of the experience as a whole, including features and associations beyond the immediate targets of goal pursuit (Murty & Adcock, 2017). Critically, both accounts predict an agency benefit for information under a participant's control but diverge for information outside that control. Our design therefore hinged on measures that participants' actions did not directly shape. Across both experiments, agency enhanced mnemonic discrimination and object description recall, and in Experiment 2 this benefit extended to room descriptions that participants never acted upon, supporting the view that agency shapes memory beyond the information directly controlled.

To our knowledge, our study provides the first demonstration that agency during encoding enhances performance on the Mnemonic Similarity Task. The MST has become a predominant behavioral assay of mnemonic discrimination in humans (Stark et al., 2013, 2019; Yassa & Stark, 2011), and a large body of work has established its sensitivity to hippocampal function. Discriminating similar lures from studied targets is selectively impaired in amnesic patients with damage limited to the hippocampus, even when basic recognition is intact (Kirwan et al., 2012), and lure discrimination declines with aging and in conditions associated with hippocampal dysfunction, including amnesic mild cognitive impairment (Stark et al., 2013; Yassa et al., 2010). Neuroimaging studies using the MST have further localized lure-related signals to the dentate gyrus and CA3 subfields (Baker et al., 2008; Berron et al., 2016; Lacy et al., 2011), consistent with computational accounts in which pattern separation reduces overlap among similar inputs (McClelland et al., 1995; Norman & O'Reilly, 2003; Treves & Rolls, 1994).

Much of this literature has treated mnemonic discrimination as an index of hippocampal integrity, comparing performance across populations, age groups, or stimulus properties (Delarazan et al., 2023; Leal & Yassa, 2018; Reagh et al., 2018; Stark et al., 2019). Far fewer

studies have asked whether the state of the individual during encoding alters the fidelity of the representations available for later discrimination (though, see Clemenson & Stark, 2015). Our findings show that agency is one such factor. Across both experiments, the same objects were remembered with greater fidelity when encountered during active control than passive observation. Importantly, placing an object required only identifying it well enough to select its location. That is, participants never acted upon the perceptual features that later distinguished that object from a similar lure. An agency benefit at this level of mnemonic detail is therefore difficult to explain through preferential strengthening of controlled information alone.

One possibility is that agency enhances the detail of item representations formed during encoding. When a perceptually similar lure is later encountered, these more detailed representations may be better able to resist interference and support discrimination between the lure and the original item. The resulting improvement is consistent with hippocampal pattern separation processes, but the present behavioral results cannot determine whether agency directly altered these processes. Additionally, lure discrimination can also be influenced by the quality of item representations outside the hippocampus and by retrieval-stage decision processes (Leal & Yassa, 2018). Moreover, the benefits of agency were not specific to mnemonic discrimination but extended across several aspects of memory, suggesting a broader influence on encoding. Thus, although the MST results demonstrate an effect of agency on mnemonic fidelity, identifying the neural processes responsible for this benefit will require direct neural measures.

Interestingly, the object-spatial placement results differed across our two experiments, despite both accounts predicting an agency benefit on this measure. Agency did not benefit object-spatial placement memory in Experiment 1, when participants were required to reproduce the precise location of each object without retrieval support. In Experiment 2, we altered the format

of retrieval trials, reducing retrieval demands by asking participants to select among three possible locations, and under these conditions a reliable agency benefit emerged. This pattern could suggest that the more demanding test used in Experiment 1 may have been less sensitive to condition differences in spatial memory. Alternatively, participants may not have encoded precise locations at all, but rather coarse, “gist” level locations that were better indexed by the recognition task. However, given that the experiments differed in other respects (i.e., between- versus within-subjects design, in-person versus online samples, number of encoded rooms and objects), we cannot attribute the emergence of the object-spatial placement benefit to retrieval demands alone. Importantly, Experiment 2 demonstrates that agency can enhance memory for the spatial information participants directly controlled, alongside its broader benefits for information outside that control.

A persistent concern in studies of agency is that active participants may simply be more engaged than passive observers, and that greater engagement, rather than agency itself, drives the memory benefit (Chrastil & Warren, 2012; Markant et al., 2014). Experiment 1 could not address this possibility because attention was assessed only once, retrospectively, at the end of the session. Experiment 2 therefore collected attention ratings immediately after each encoding block, providing condition-specific measures of engagement closer in time to encoding. Every agency effect remained unaffected when taking these ratings into consideration, and attention did not independently predict performance on any measure. It is important to note that these analyses do not establish that attention plays no role in the effects of agency. Self-reported attention provides only a coarse measure of engagement and may fail to capture differences in moment-to-moment allocation of attention. Moreover, to the extent that agency itself changes how attention is deployed, attentional processes could form part of the pathway through which control influences

memory. Nevertheless, the present findings argue against a simple account in which the benefits observed here are reducible to Active participants reporting greater attentional engagement during encoding.

The broad memory benefits observed here may arise because agency changes the motivational state in which encoding occurs. One framework that anticipates such a pattern distinguishes between motivational states with different mnemonic consequences (Murty & Adcock, 2017). Interrogative states, characterized by exploration and the active resolution of uncertainty, are proposed to engage dopaminergic projections from the ventral tegmental area to the hippocampus and support rich, relational encoding. Imperative states, driven by salient and proximal goals, are instead associated with noradrenergic and amygdala-cortical circuitry that favors sparser, item-centered memories (Clewett & Murty, 2019; Murty & Adcock, 2017). This framework builds on broader evidence that dopaminergic input to the hippocampus modulates the encoding and stabilization of new memories (Lisman & Grace, 2005; Shohamy & Adcock, 2010).

Viewed through this framework, the broad pattern in Experiment 2 resembles the relational encoding attributed to interrogative states. Agency enhanced memory for item-context associations, perceptual details supporting lure discrimination, and descriptive information that was incidental to participants' actions, consistent with the possibility that active control promoted a broader encoding state in which information beyond the immediate target of action was incorporated into memory. Whether the same state was engaged in Experiment 1 is less clear. Agency produced comparable benefits for object recognition, mnemonic discrimination, and object-description recall across experiments, but the object-spatial placement measure in Experiment 1 may not have been sufficiently sensitive to determine whether item-context associations were similarly enhanced. At the same time, the framework allows for the possibility

that task demands shift encoding toward a more item-centered mode. Our data cannot resolve this difference across experiments, and resolving this question will require measures that index encoding state more directly than behavior alone. High-resolution fMRI could test whether agency modulates dentate gyrus and CA3 responses during encoding, whether these responses change with mesolimbic engagement, and whether their expression varies with task demands in the manner predicted by the interrogative-imperative distinction (Adcock et al., 2006; Murty et al., 2015).

Future work should also isolate which aspects of active control are responsible for these memory benefits. In the present experiments, the Active and Passive conditions differed not only in volitional control but also in motor activity. Active participants physically dragged and dropped objects, whereas Passive participants did not. Memory benefits of physical performance are well documented in the literature on enactment effects, in which performing an action produces better memory than watching or reading about it (Cohen, 1981; Engelkamp, 1991). We therefore cannot fully dissociate the contributions of volitional control from those of motor engagement in the present design. At the same time, the benefits of Active encoding extended to information that was not itself the target of any motor action, including the perceptual details supporting lure discrimination and, in Experiment 2, descriptions of the rooms. Future work could separate these factors directly by requiring participants to execute computer-determined placements, preserving motor activity while removing volitional control.

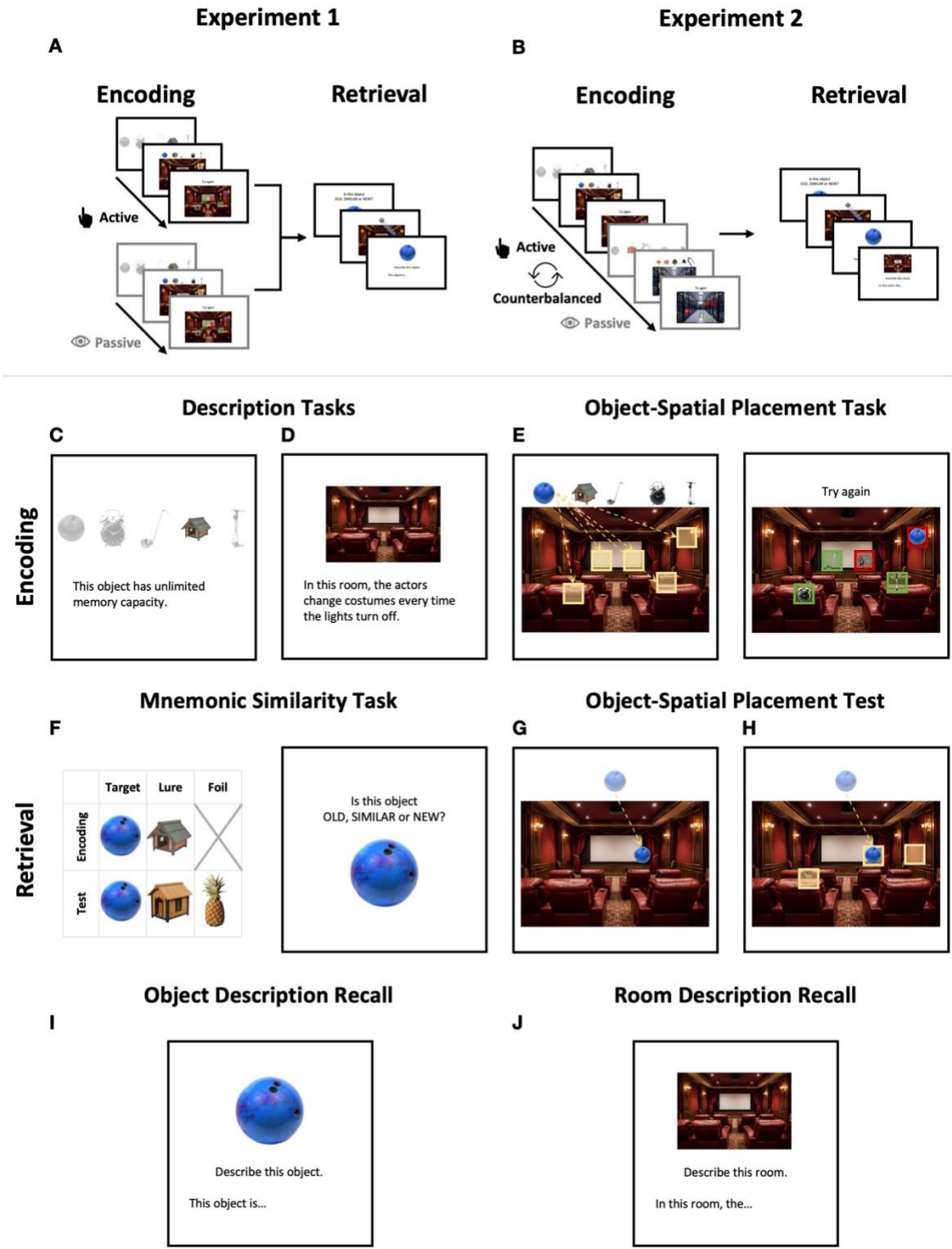
Despite these remaining questions, the present findings show that the benefits of taking control over one's experience are not confined to the information one controls. Agency enhanced the fidelity of object representations, memory for object-spatial location associations, and recall of descriptive information that was incidental to participants' actions. These effects emerged across

1 measures that placed different demands on memory and, in Experiment 2, remained after
2 accounting for self-reported attention. Additionally, this pattern may be supported by hippocampal
3 processes of pattern separation and relational binding, though direct neural measures are still
4 needed to determine whether agency modulates this directly. Together, these findings suggest that
5 agency enhances memory beyond the information directly controlled, extending to both the details
6 of individual items and the broader context in which they were encountered.

7 Broadly, this work provides a way to examine how control over an unfolding experience
8 influences multiple aspects of memory within the same event. The findings also resonate with a
9 long-standing principle underlying active learning in which making choices, testing possibilities,
10 and correcting errors may influence memory beyond the immediate products of those actions. Our
11 results suggest that exerting control may broaden what is encoded, strengthening not only the goal-
12 relevant information but also other features encountered along the way.

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5. FIGURES



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Figure 1. Methods: Experimental Design, Procedure, and Task Structure. (A) Overview of Experiment 1 Design. Using a between-subjects design, participants in the Active condition played the Visuospatial Game, placing objects into their correct

locations through trial and error, whereas participants in the Passive condition viewed a yoked replay of an Active participant's gameplay. Both conditions then completed an identical Retrieval Phase. **(B) Overview of Experiment 2 Design.** Using a within-subjects design, all participants completed both Active and Passive blocks during encoding, presented in alternating blocks of four rooms with the starting condition counterbalanced across participants. **(C) Object Description Task.** Within each room, participants learned a description of each object's special capability. **(D) Room Description Task (Experiment 2 Only).** Participants were shown an image of the room and learned a description of the room's special capability. **(E) Object-Spatial Placement Task.** Objects appeared in randomized locations within each room alongside predetermined target locations (left). Participants dragged each object to a target location and received feedback indicating whether each placement was correct, repeating the trial until all objects were correctly placed (right; green = correct, red = incorrect). **(F) Mnemonic Similarity Task.** At test, participants viewed exact repetitions of studied objects (Targets), perceptually similar objects not previously seen (Lures), and novel objects (Foils), and judged whether each was "Old," "Similar," or "New." **(G) Object-Spatial Placement Test (Experiment 1).** Participants selected the room in which the object had appeared and then dragged the object to its remembered location anywhere within the room. **(H) Object-Spatial Placement Test (Experiment 2).** Participants were shown the room along with three markers indicating the object's possible target locations and dragged the object to the marker they believed was correct, reducing the precision demands of retrieval as a recognition-based task. **(I) Object Description Recall.** Participants typed what they remembered about each object's special capability. **(J) Room Description Recall (Experiment 2 Only).** Participants typed what they remembered about each room's special capability.

Experiment 1

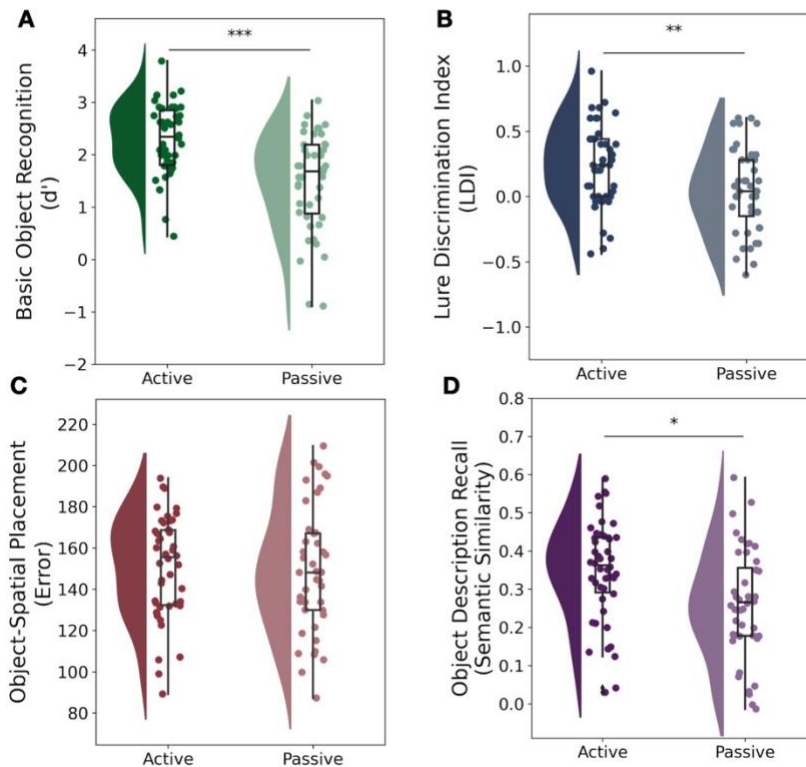


Figure 2. Experiment 1 Results. (A) Basic object recognition (d') from the Mnemonic Similarity Task, indexing discrimination of studied Targets from novel Foils. (B) Lure Discrimination Index (LDI), indexing discrimination of studied Targets from perceptually similar Lures. (C) Object-Spatial Placement Error, calculated as the Euclidean distance in pixels between each placement and the correct target location; lower values indicate more accurate placement. (D) Object Description Recall, scored as the cosine similarity between each participant's recalled description and the original description; higher values indicate greater semantic overlap. Raincloud plots show the distribution (half-violin), individual participant scores (points), and boxplot summaries for Active (dark) and Passive (light) participants. Boxplots show the median and interquartile range, with whiskers extending to $1.5 \times \text{IQR}$. **Key:** * $p < .05$, ** $p < .01$, *** $p < .001$.

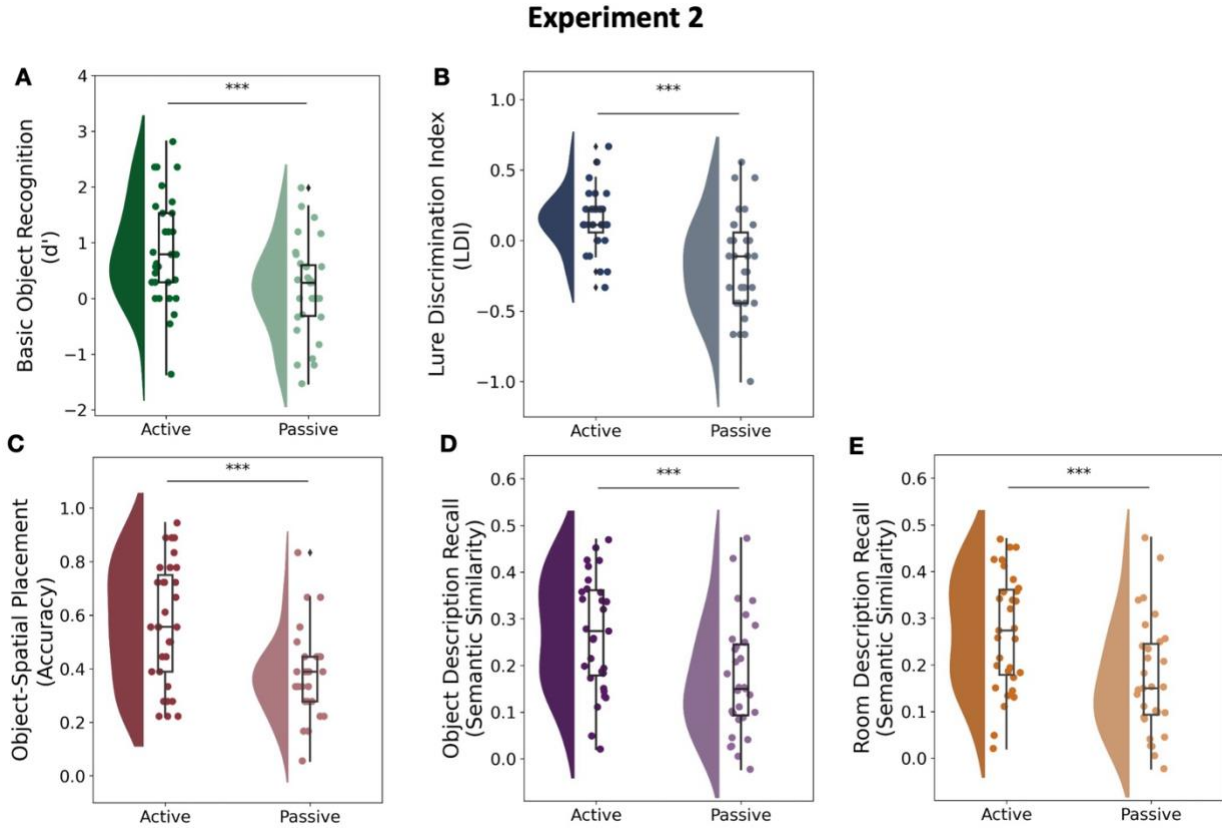


Figure 3. Experiment 2 Results. (A) **Basic object recognition** (d') from the Mnemonic Similarity Task, indexing discrimination of studied Targets from novel Foils. (B) **Lure Discrimination Index (LDI)**, indexing discrimination of studied Targets from perceptually similar Lures. (C) **Object-Spatial Placement Accuracy**, calculated as the proportion of objects correctly placed among three possible target locations; higher values indicate more accurate placement. (D) **Object Description Recall**, scored as the cosine similarity between each participant's recalled description and the original description; higher values indicate greater semantic overlap. (E) **Room Description Recall**, scored as the cosine similarity between each participant's recalled description and the original description; higher values indicate greater semantic overlap. Raincloud plots show the distribution (half-violin), individual participant scores (points), and boxplot summaries for Active (dark) and Passive (light) participants. Boxplots show the median and interquartile range, with whiskers extending to $1.5 \times IQR$. **Key:** $*p < .05$, $***p < .001$.

6. TABLES

Table 1. Experiment 1 Post-Task Questionnaire. Items were rated on a 0-6 scale anchored by two statements, with 0 corresponding to the first anchor listed and 6 to the second. **Presence:** “How present did you feel during the game task?” (0 = I felt like I was there in the room, not just on the screen; 6 = I felt very detached from the game). **Attention:** “How attentive did you feel to the game task?” (0 = I felt “in the zone”; 6 = I felt unable to pay attention to the game). **Encoding Confidence:** “How well could you remember information during the game task?” (0 = I would easily remember what was happening in the game; 6 = I would constantly forget what was happening in the game). **Retrieval Confidence:** “How well could you recall information in the memory tests after the game task?” (0 = It felt easy to recall all the information prompted in the memory tests; 6 = It felt impossible to recall any information prompted in the memory tests). **Control:** “How in control did you feel during the game task?” (0 = I felt like I had full control over the game task; 6 = I felt like I had no control over the game task). **Randomness:** “How easily did you realize the correct item placements during the game task?” (0 = I felt like the correct placements “clicked” to me; 6 = I felt like the correct placements were random matches). **Encoding Strategy:** “How did you identify the correct placements of the items during the game task?” (0 = relative to the placement of other items; 6 = individually, irrespective of the placement of other items). **Retrieval Strategy:** “How did you recall the correct placements of the items in the memory tests after the game task?” (0 = relative to the placement of other items; 6 = individually, irrespective of the placement of other items). **Perspective:** “Did you have a precise or holistic perspective of where the items belonged in the scene?” (0 = I viewed the objects specific to their precise location; 6 = I viewed all the objects as within the larger, holistic scene). Lower values therefore indicate greater presence, attention, confidence, and control. Rows are ordered by *p*-value (*p*).

Experiment 1 Post-Task Questionnaire				
Item	Active <i>M (SD)</i>	Passive <i>M (SD)</i>	<i>t</i>	<i>p</i>
Control	2.40 (1.50)	4.74 (1.45)	-7.39	<.001
Attention	2.29 (1.29)	3.86 (1.17)	-5.98	<.001
Randomness	3.17 (1.90)	4.52 (1.59)	-3.64	<.001
Encoding Confidence	2.83 (1.69)	3.60 (1.64)	-2.21	.030
Presence	3.28 (1.67)	3.91 (1.46)	-1.89	.062
Perspective	3.15 (1.78)	3.30 (1.81)	-0.38	.706
Encoding Strategy	2.63 (2.11)	2.48 (1.53)	0.39	.696
Retrieval Strategy	3.50 (1.62)	3.63 (1.72)	-0.36	.718
Retrieval Confidence	4.15 (1.11)	4.23 (1.01)	-0.33	.739

Table 2. Experiment 2 Post-Task Questionnaire. Items were rated on a 0-6 scale anchored by two statements, with 0 corresponding to the first anchor listed and 6 to the second. **Encoding Confidence:** “How well could you remember information when playing/watching the game task?” (0 = I could easily remember what was happening in the game; 6 = I felt very detached from the game). **Control:** “How in control did you feel while playing/watching the game task?” (0 = I felt like I had full control over the game task; 6 = I felt like I had no control over the game task). Lower values therefore indicate greater encoding confidence and control.

Experiment 2 Post-Task Questionnaire				
Item	Active <i>M (SD)</i>	Passive <i>M (SD)</i>	<i>t</i>	<i>p</i>
Control	1.94 (1.93)	4.35 (2.07)	-5.30	<.001
Encoding Confidence	2.71 (1.66)	4.74 (1.36)	-4.21	<.001

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